Method Development for Proteomic Host Profiling of Adult Patients with Suspected Infection and Sepsis

UC DAVIS
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Project Overview

A: 40 remnant plasma samples from adults (age >= 18) with normal and abnormal procalcitonin levels collected.
B: Laboratory protocol optimized.
C: Samples extracted w/ Zip Tips, mixed w/ matrix, plated.
D: MALDI-TOF MS yields MS profile.
E: Peak lists obtained.
F: Data Analysis.
G: Interesting peaks/peak patterns identified.
H: Potential sepsis biomarkers.

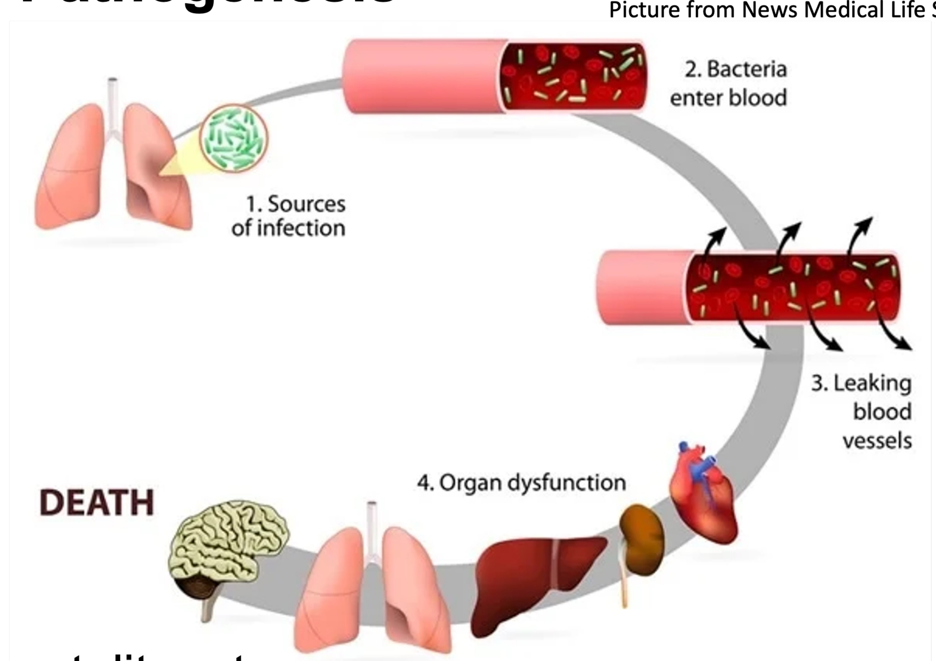
Mass spectrum profiles obtained using the Bruker ultrafleXtreme MALDI-TOF mass spectrometer; Mass spectra processing using FlexAnalysis; Mass peak lists per sample were fed to MassUp ML software.

Picture made using Biorender.

Background

Sepsis Pathogenesis

Picture from News Medical Life Sciences.



- High mortality rate.
- No effective early diagnosis method.
- Biomarker is a promising field for studies on early sepsis detection.
- Abnormal procalcitonin indicates high likelihood for sepsis.

Matrix Assisted Laser Desorption Ionization-Time of Flight (MALDI-TOF) Mass Spectrometry

MALDI | TOF

Plasma samples & CHCA Matrix hit with laser, ionizes & vaporizes.

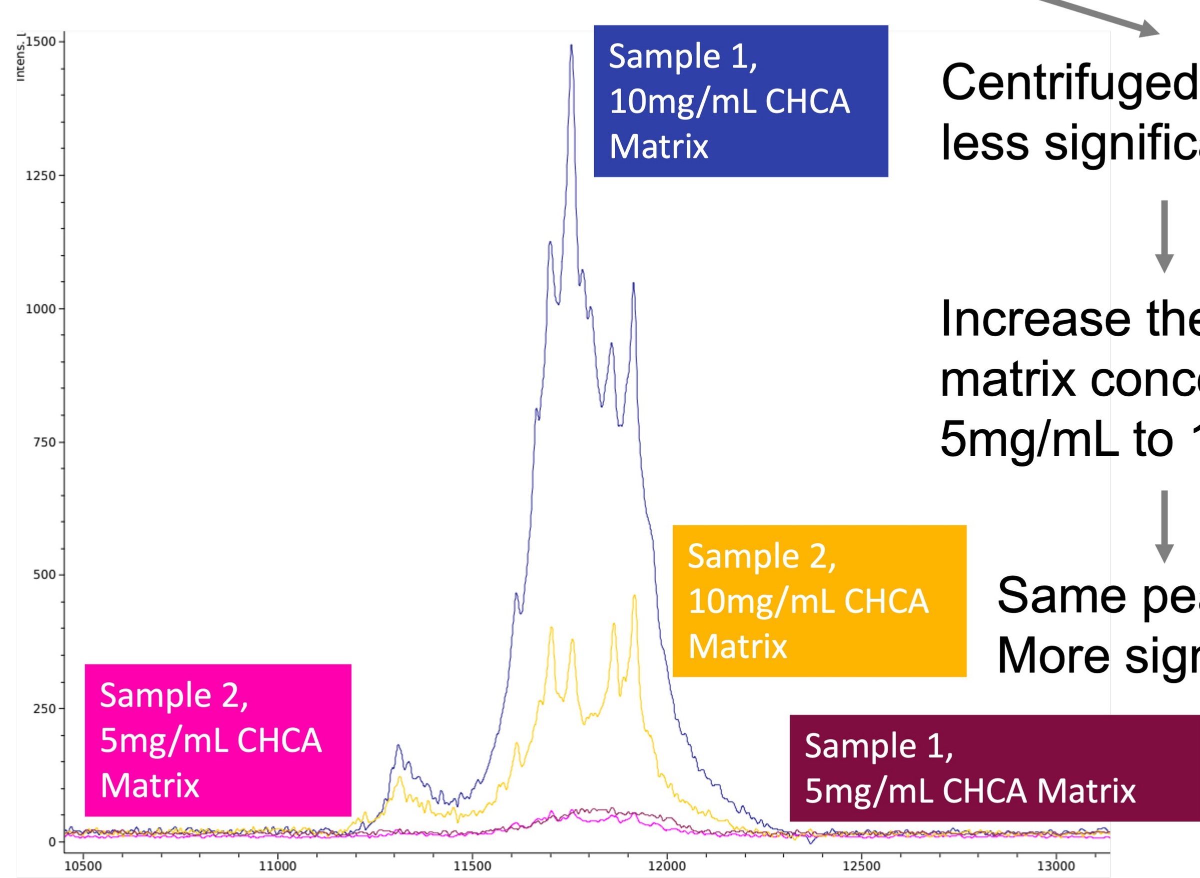
Ions go through vacuum tube.

Time of flight measured.
Smaller Mass→ Shorter Time, vice versa.

Picture from Bruker.

Protocol Optimization

Uncentrifuged plasma impact on Zip Tips

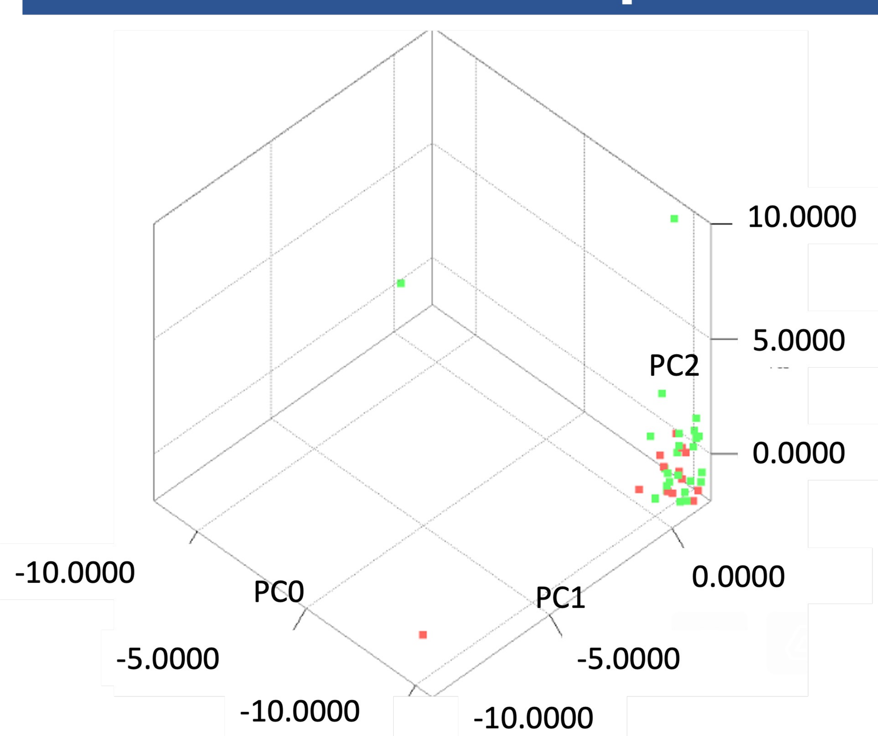


Centrifuged samples had less significant peaks.

Increase the CHCA matrix concentration from 5mg/mL to 10mg/mL.

Same peak captured. More significance.

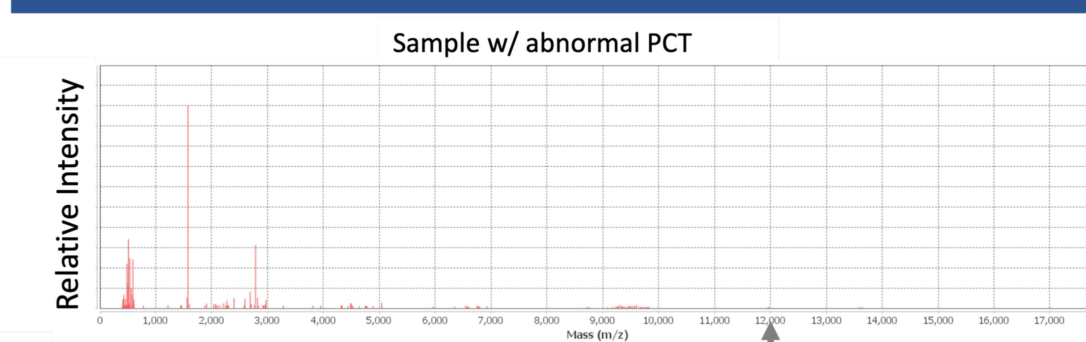
Results: Principal Component Analysis

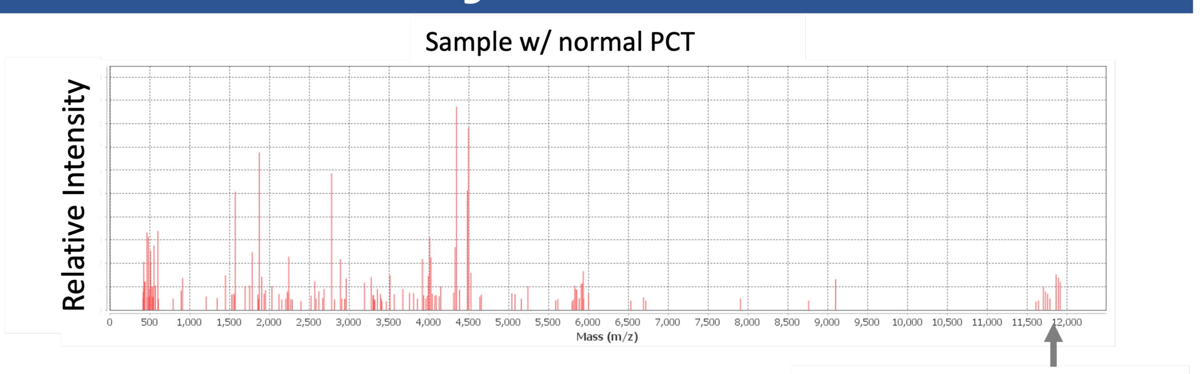


Green: samples w/ normal PCT Results.
Red: samples w/ abnormal PCT results.

Two clusters intermixed together:
--> No clear distinction.
--> More samples may be needed.

Results: Biomarker Discovery





m/z ~= 12,000, no peak

m/z ~= 12,000, has peak

	2586.68	12022.23
Normal PCT value	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0.013502	0
	0	0
Abnormal PCT value	0.054624	0
	0.012442	0
	0.012442	0
	0	0.180521
	0.025834	0
	0.008923	0
	0.03128	0
	0	0.003109
	0	0
	0	0
	0	0.023661
	0.012467	0
	0.012415	0
	0	0
	0	0.004976
	0	0.004539
	0	0
	0.005635	0
	0	0
	0	0
	0	0.003188
	0.005347	0
	0	0.003521
	0	0.006888
	0.03197	0
	0.363077	0.32

Mass/Charge Ratio of a peak

The intensity of this mass/charge ratio in each sample.

The interesting peaks >30% more present in abnormal PCT samples.

1. Mass spectra obtained.
2. Peak lists extracted.
3. Peak lists aligned across samples.
4. Interesting peaks identified.

Conclusions

- MALDI-TOF MS analysis of plasma samples requires optimization when used with Zip Tips.
- Principal Component Analysis showed no distinct separation, showing that sepsis and inflammation are very heterogeneous.
- However, several interesting peaks have been identified, which appeared more often in patients with abnormal procalcitonin levels.
- An increased sample size is necessary to further differentiate sepsis from non-sepsis and allow for the application of other analytical techniques, such as machine learning, for analysis.